MILITARY SPECIFICATION

SNOWPLOW, TRUCK MOUNTING, POWER REVERSIBLE BLADE MXU-512/E

1. SCOPE

1.1 This specification covers one type of hydraulically operated reversible blade type snowplow, designated MXU-512/E.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-S-781	Steel Strapping, Flat
TT-V-121	Varnish, Spar, Water-Resisting
UU-T-81	Tags, Shipping and Stock
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-C-650	Crates, Wood, Open and Covered
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-C-104	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-F-3541	Fittings, Lubrication
MIL-C-3774	Crates, Wood, Open, 12,000- and 16,000-Pound Capacity
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-H-13531	Hose, Rubber and Hose Assembly, Rubber (Hydraulic, Flexible)
MIL-V-13811	Varnish, Waterproofing, Electrical, Ignition
MIL-H-15424	Hand Tools; Packaging of
MIL-T-83118	Trucks, Snow Removal Equipment A/S32U-35 and A/S32U-36 (Heavy Duty)

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STANDARDS

Federal

FED-STD-151	Metals, Test Methods

Military

MIL-STD-12	Abbreviations for use on Drawings and in Technical-Type Publications
MIL-STD-17	Mechanical Symbols (Other Than Aeronautical, Aerospace, and Spacecraft)
MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-209	Slinging Eyes and Attachments for Lifting and Tying Down Heavy Military Equipment
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equippment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods
MS24333	Coupling Assembly, Hydraulic, Self-Sealing, Quick Disconnect, Flared Fitting to Internal Thread Boss
MS24334	Coupling Assembly, Hydraulic, Self-Sealing, Flareless Fitting to Internal Thread Boss

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Society of Automotive Engineers, Incorporated

SAE Standards and Recommended Practices

J739	Grader Cutting Edge
J740	Countersunk Square Holes for Cutting Edges and End Bits

(Application for copies should be addressed to the Society of Automotive Engineers, Incorporated, Two Pennsylvania Plaza, New York, New York 10001.)

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United States of America Standards

B18.9 Plow Bolts

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The snowplow shall consist of the following components:

Item Name	See Requirement
Frame	3.6.1
Moldboard	3.6.2
Hydraulic components	3.6.3.

3.3 Materials. Materials shall be as specified herein. Materials not definitely specified shall be of the best commercial quality and entirely suitable for the purpose. Materials shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.4 Design and construction. The snowplow shall be designed and constructed as a straight, power reversible blade assembly for installation on and functioning with snow removal equipment trucks in accordance with MIL-T-83118 equipped with the specified front hitch. All parts necessary for installation and removal of the snowplow and for efficient and satisfactory operation shall be furnished. The time required to install or remove the snowplow from the truck shall not exceed 15 minutes. All parts of the snowplow shall be of such size, materials, and strength as to withstand the full output power of the truck in all gear ratios and under all operating conditions specified herein. In all positions of the moldboard, the snowplow shall operate without blade chatter, weaving, rocking, or other actions not under complete control of the operator.

3.4.1 Snow clearance. The snowplow shall be so designed that when mounted on the truck and operating on a smooth level surface at a continuous forward speed of 30 mph and locked at a 40° angle, the snowplow will clear a swath not less than 13 feet 6 inches wide in snow 3 inches deep weighing 10 pcf or equivalent depth density ratio. The residue left on the surface shall not exceed 1/2 inch in depth.

3.4.2 The snowplow shall be so designed and constructed that parts will not work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, and service.

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3.4.3 Center of gravity. The distance from the hitch to the center of gravity of the snowplow shall not exceed 40 inches.

3.4.4 Road clearance. When in the raised position, the snowplow shall have a minimum ground clearance of 10 inches.

3.4.5 Reliability. The snowplow shall be designed and constructed as specified herein and to successfully complete the tests specified in 4.4.3 with no failure preventing satisfactory performance.

3.4.6 Maintainability. The snowplow shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with reliability and performance specified herein

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure

c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessiblity

d. Permit maintenance with general-purpose tools and equipment normally available commercially. The use of special-purpose tools and equipment shall be subject to approval by the procuring activity

e. Minimum number of tools required for maintenance by design practices such as reducing the variety of bolt head sizes to the practicable minimum.

3.4.7 Foolproofness. Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided where practical. That mounting shall allow the item to be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

3.4.8 Lifting rings. Lifting rings shall be provided to enable the snowplow to be lifted in its normal position. All lifting rings shall be designed and located in accordance with MIL-STD-209.

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3.5 Performance. The snowplow, when installed on an A/S32U-35 snow removal equipment truck in accordance with MIL-T-83118, ballasted to an operating weight of 42,000 pounds (with the attachment) shall withstand the following conditions without detrimental effect to subsequent operation:

3.5.1 Operation. The snowplow shall operate satisfactorily as follows:

a. Demonstrate snow clearance in accordance with 3.4.1 for not less than 8 hours at full output power of the truck

b. Reverse from full cast in one direction to full cast in the opposite direction in not less than 15 seconds nor more than 30 seconds

c. With the safety mechanism manually locked-out, the snowplow shall withstand in bulldoze position and in all combinations of plow widths (center third, right two-thirds, left two-thirds, and full width) an evenly distributed force of 25,000 pounds distributed as follows:

(1) The bottom one-third of the plow face

(2) The bottom two-thirds of the plow face

(3) The entire plow face

The force shall be applied in a direction parallel to the longitudinal centerline of the truck.

3.5.2 Environmental conditions

3.5.2.1 Operating temperatures. The snowplow shall operate satisfactorily in ambient temperatures ranging from -40° to +40°F.

3.5.2.2 Storage temperatures. The snowplow shall withstand storage at any temperature ranging from -80° to +160°F.

3.6 Details of components

3.6.1 Frame. The frame shall be fabricated of structural material of sufficient strength to support the moldboard and accessories and to withstand plowing stresses. Plowing stresses shall be transmitted to the hitch. Ample clearance between the plow and the hitch shall be provided for the specified plow movements.

3.6.2 Moldboard. The moldboard shall be constructed of aluminum of adequate thickness and shall be correctly curved for efficient displacement of snow. It shall be sufficiently reinforced to withstand the loadings specified in 3.5.1. The moldboard shall be not less than 18 feet wide and 40 inches high as measured over the blade cutting edge. The moldboard shall be shaped and drilled to permit installation of the blade cutting edge.

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3.6.2.1 Blade cutting edge. The blade cutting edge shall conform to SAE Standard J739 for grader cutting edges, except it shall be 3/4 inch thick, at least 6 inches high, and with the bolt holes located 4 inches from the cutting edge. The blade cutting edge shall be in sections 3- and 4-feet long to permit interchanging of blade sections to compensate for unevenness of wear. The bolt holes shall be in accordance with SAE Standard J740. Number 3 head plow bolts, grade 8, in accordance with USA Standard B18.9 with locknut and lockwasher shall be used. The blade shall be constructed of C1015 to C1025 steel, shall be removable, and shall fit the moldboard. The cutting edge in contact with the pavement shall have tungsten carbide inserts brazed into the blade for the full length of the blade. The inserts shall be brazed on the front and back sides, full depth, without voids. The inserts shalll have a nominal hardness of Rockwell A 88.5. After brazing, the blades shall be stress relieved as necessary to eliminate tension in the inserts. The difference between the highest and lowest tip point shall not exceed 1/32 inch in any blade section. The cutting edge, with the inserts, shall be capable of at least 12,000 miles of operation in continuous contact with concrete pavement.

3.6.2.2 Deflector. The moldboard shall be equipped with a deflector extending the full length of the moldboard. When plowing 3 inches of snow at a density of 5 pcf at maximum plowing angle, at a vehicle speed of 30 mph, the deflector shall prevent snow from topping the plow.

3.6.3 Hydraulic components. The hydraulic components shall consist of rams, hoses, piping, fittings, and all other parts essential to the practical and efficient operation of the snowplow. These components shall be compatible with the control system of the snow removal equipment trucks. A minimum number of fittings, joints, and connections shall be used to prevent undue back pressure, clogging, and leakage. All parts shall be mounted to prevent detrimental vibration and shock. Quick-disconnect fittings shall be provided to permit connecting the hydraulic feed lines to a panel which has fittings conforming to MS24333-8 or MS24334-8.

3.6.3.1 Flexible hose and hydraulic tubing. Hose for hydraulic lines shall conform to MIL-H-13531, type II, class A. The hydraulic tubing and flexible hose shall be of adequate size to permit free flow of hydraulic fluid conforming to MIL-H-5606 at the temperatures specified in 3.5.2.

3.6.3.2 Hydraulic rams. Rams and interior cylinder surfaces shall be plated in accordance with QQ-C-320, type II.

3.6.4 Rotating mechanism. The moldboard shall be hydraulically adjustable in the horizontal plane through plowing angles up to 40° in either direction. The snowplow shall be so designed that the moldboard can be maintained in the 0°, 20°, 35°, and 40° positions by the driver in the cab.

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3.6.5 Safety mechanism. A safety mechanism, so designed that it can be manually locked out without the use of tools, which will allow the plow to ease over obstructions at least 3 inches high and then automatically return the plow to its proper operating position shall be provided. The blade section safety mechanisms shall function whenever the plowing stresses exceed 80 percent of the maximum stresses developed when the plow is loaded as in 3.5.1.c.

3.7 Lubrication. All moving parts shall be provided with suitable means of lubrication and shall operate satisfactorily when lubricated with standard military lubricants as specified in MIL-M-8090.

3.7.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541 and shall be located in accessible, protected positions.

3.7.2 Lubrication chart. A lubrication chart shall be provided directing attention to all lubrication fittings and shall specify the range and grade of lubricant required for critical temperatures. The chart shall be permanently attached in a protected, conspicuous location.

3.7.3 Initial service. The snowplow shall be completely serviced prior to delivery.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part number and changes thereto.

3.9 Weight. The weight of the snowplow shall be the minimum consistent with the requirements specified herein and shall not exceed 3,000 pounds.

3.10 Finishes and protective coatings. The snowplow parts normally painted shall be painted in accordance with the manufacturer's commercial practice. The color shall be yellow, designation DG, conforming to MIL-STD-808, except the portion of the moldboard facing the driver shall be lusterless black, designation UL.

3.11 Markings

3.11.1 Data plates. Transportation data plates conforming to MIL-P-514 shall be provided. All marking plates, placards, charts, instruction plates, et cetera, shall conform to MIL-P-514. Size and shape shall be consistent with information required thereon. Abbreviations shall be in accordance with MIL-STD-12. Symbols shall be in accordance with MIL-STD-17.

3.12 Identification product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.13 Government-loaned property. When specified in the contract (see 6.2), the Government will load one A/S32U-35 snow removal equipment conforming to MIL-T-83118 to the contractor upon his request.

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3.14 Workmanship. Workmanship shall be of the highest grade throughout and in accordance with good commercial practice for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the snowplow shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Mounting. The snowplow shall be mounted on a A/S32U-35 snow removal equipment truck conforming to MIL-T-83118 for the tests specified in 4.6.3 and 4.6.5. The truck shall be ballasted to an operating weight of 42,000 pounds (with the attachment). For all other tests, a fixture, in lieu of the truck, may be used to simulate operating characteristics. The fixture shall duplicate the snowplow hitch, controls, pressures, and hydraulic action required to operate all operating components.

4.3.2 Preparation for test. Prior to testing, the snowplow shall be serviced with lubricants in accordance with the instructions of the manufacturer.

4.4 Preproduction testing (see 6.2)

4.4.1 Test sample. One snowplow shall be subjected to the tests specified in 4.4.3.

4.4.2 Test report. After completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded

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b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item

f. Test activity or contractor comments on use or maintenance conditions to be avoided to increase the reliability or useful life of the item

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.4.3 Preproduction tests. The preproduction tests shall consist of all tests specified under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual test

b. Sampling test.

4.5.1 Individual test. Each snowplow shall be subjected to an examination of product test as described under 4.6.

4.5.2 Sampling test

4.5.2.1 Periodic sampling. One snowplow shall be selected at random from each 25 or fraction thereof produced and subjected to the test specified in 4.6.2 except it shall be reversed only 10 times.

4.5.2.1.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

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4.5.2.1.2 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a test failure. Final acceptance of the items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The snowplow shall be visually examined for compliance with the requirements of this specification with regard to completeness of assembly, presence of all components and accessories, quality of workmanship and materials, dimensions, hydraulic component leaks, marking, mounting suitability and simplicity, and the functional operation of all components, assemblies, accessories, and controls.

4.6.2 Mounting and functioning test. The snowplow shall be mounted on the truck or fixture as specified in 4.3.1 and all dimensions, fitting locations, and operating clearances shall be checked. All operable components shall be operated to determine that they function satisfactorily. The snowplow shall be reversed from full cast in one direction to full cast in the opposite direction 100 times. The snowplow shall reverse from full right to full left in not less than 15 seconds nor more than 30 seconds.

4.6.3 Strength test. When mounted in accordance with 4.3.1, an evenly distributed 25,000-pound force shall be applied to the face of the snowplow as specified in 3.5.1. There shall be no impairment of function when subjected to this force nor permanent deformation of any parts after removal of this force.

4.6.3.1 Safety mechanism test. The stresses developed before the safety mechanism functions shall be determined and shall be in accordance with 3.6.5.

4.6.4 Hardness test. The tungsten carbide inserts shall be subjected to a hardness test in accordance with FED-STD-151 to determine compliance with hardness requirements of 3.6.2.1.

4.6.5 Operational test. The truck-mounted snowplow shall be operated to determine that it functions satisfactorily as specified herein. Actual plowing operations shall be accomplished in snow for a minimum of 8 hours to determine that the various components are of adequate strength and function properly. Snow shall be not less than 3 ±1 inch average depth and weigh 5 to 20 pcf. Snow clearance shall be in accordance with 3.4.1. There shall be no loss or fracturing of any tungsten carbide inserts. Residual snow shall not exceed 1/2 inch in depth. No evidence of hydraulic leaking or component failure shall occur.

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4.6.5.1 Installation and removal test. On completion of the 8-hour operational test, the snowplow shall be removed from the truck. The truck shall be moved out of the immediate area, maneuvered through 360° and approach the snowplow. The plow shall be installed on the truck. Removal or installation shall not exceed 15 minutes.

4.6.5.2 Endurance test. Prior to actual endurance test initiation, the snowplow shall be operated for a break-in period sufficient to obtain uniform wear over the full width of the tungsten carbide cutting edge. Accurate measurements of at least three tungsten carbide tips per blade section shall be obtained, recorded, and the measured tips marked or otherwise identified for future reference. The snowplow shall then be driven a distance of 1,500 miles with the blade cutting edge in continuous contact with concrete surface. Approximately 1,400 miles shall be with the plow angled 40° (700 miles right and 700 miles left) and 100 miles in bulldoze position. Lack of snow shall not be cause for delay of this test. On completion of this test, all of the previously identified tips shall be accurately remeasured and compared with the original data. The carbide tips shall not exhibit wear in excess of one-eighth of the available wearing surface. Blade sections may be interchanged (but not replaced) as required during this test to assure evenness of wear.

4.6.6 Center of gravity. The location of the center of gravity shall be determined and shall be in compliance with that specified in 3.4.3.

4.6.7 Blade section interchangeability test. All 3- and 4-foot blade sections shall be interchanged to assure ability to compensate for uneveness of wear.

4.6.8 Weight. The weight of the snowplow assembly specified herein shall be determined and shall not exceed that specified in 3.9.

4.6.9 Environmental tests. The snowplow, mounted in accordance with 4.3.1, shall be subjected to the following environmental tests in accordance with MIL-STD-810 for storage from -80° to +160°F and operation from -40° to +40°F. After stabilizing at operating temperature, 100 cycles of the following operations shall be performed:

a. Raise and lower the snowplow from a full down to a full up position and vice versa

b. Rotate the snowplow in the horizontal plane from the extreme right to the extreme left position

c. Engage the locking mechanism at each stop position

During or after the tests, failure of the rams to fully raise or lower connected components, failure of rotating mechanism to smoothly rotate from one cast position to another, any evidence of hydraulic leakage or component failure, or inability to accomplish the required performance shall be cause for rejection.

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4.6.9.1 High temperature. The snowplow shall be subjected to high temperature storage and operation in accordance with method 501, procedure I. After stabilizing at operating temperature, the snowplow shall be subjected to the tests specified in 4.6.9.

4.6.9.2 Low temperature. The snowplow shall be subjected to low temperature storage and operation in accordance with method 502, procedure I. The storage temperature shall be maintained for 48 hours. After stabilizing at operating temperature, the snowplow shall be subjected to the tests specified in 4.6.9.

4.6.10 Reliability demonstration and test. Satisfactory completion of all tests specified herein shall be a demonstration of compliance with the requirements of this specification.

4.6.11 Maintainability evaluation. The snowplow shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given maintenance with a minimum number of common tools, servicing, and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of dirt, snow, ice, et cetera, that may hinder servicing and operation.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Disassembly. Disassembly shall be the minimum necessary to reduce cubage and safeguard parts known to be subject to damage, pilferage, and loss. Bolts, nuts, washers, and pins shall be placed in one of the mating parts and secured to prevent their loss.

5.1.1.2 Matchmarking. When necessary to facilitate reassembly, parts that are removed and mating parts shall be matchmarked with lusterless white enamel, overcoated with clear spar varnish conforming to TT-V-121. Small parts shall have matchmarking identification on waterproof, cloth shipping tags conforming to UU-T-81, type A, attached to mating parts.

5.1.1.3 Unpainted surfaces. Unpainted metal surfaces requiring the application of scarifiers, clevises, bolts and nuts, grease fittings, hose coupling and similar parts, shall be coated with type P-1 preservative. Unpainted metal information plates shall be coated with electrical ignition water-proofing varnish conforming to MIL-V-13811.

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5.1.1.4 Machined surfaces. Unpainted exterior machined surfaces of component parts shall be coated with type P-2 or P-6 preservative. The coated surfaces shall be wrapped or covered with greaseproof barrier material conforming to MIL-B-121, secured in place with waterproof, pressure-sensitive adhesive conforming to PPP-T-60, type III, class 1.

5.1.1.5 Hydraulic systems. The hydraulic fluid supply lines shall be flushed with the approved hydraulic fluid. Wherever practical, pistons shall be fully retracted and any exposed surfaces of the piston rods shall be coated with type P-6 preservative and spirally wrapped with greaseproof barrier material conforming to MIL-B-121, secured in place with waterproof, pressure-sensitive adhesive tape conforming to PPP-T-60, type III, class 1. Open ends of hose shall be sealed with tape conforming to PPP-T-60, type III, class 1.

5.1.1.6 Operating and control levers. Unpainted control levers shall be coated with type P-1 preservative.

5.1.1.7 Extra cutting edges. Unpainted surfaces of extra cutting edges shall be coated with type P-1 preservative.

5.1.1.8 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with MIL-H-15424.

5.1.1.9 Technical publications. Technical publications shall be packaged in accordance with MIL-P-116, method 1C1-1.

5.1.1.10 Intermediate packaging. Component parts removed from each snowplow and repair parts shall be intermediate packaged in nailed wood boxes conforming to PPP-B-621, class 2, style optional, or in wood cleated-plywood boxes conforming to PPP-B-601, overseas type, style optional.

5.1.2 Level C. The snowplow, component parts, tools, and publications shall be preserved and packaged in accordance with the manufacturer's standard practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The snowplow shall be packed in exterior containers conforming to PPP-C-650, MIL-C-3774, or MIL-C-104. Packaged and unpackaged components shall be placed and secured with flat steel strapping conforming to QQ-S-781 within the shipping container to prevent movement or damage to the components or the snowplow.

5.2.2 Level B. The snowplow shall be packed as specified for level A, except the ends, sides, and top assemblies shall be omitted and the snow removal unit and component parts shall be shipped on the skid assembly of the container specification selected.

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5.2.3 Level C. The snowplow, packaged and unpackaged component parts, tools, and publications, shall be packed in compliance with Uniform Freight Classification Rules, *National Motor Freight Classification Rules*, or other carriers rules applicable to the mode of transportation.

5.2.3.1 Blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186.

5.3 Marking. In addition to any special marking required, packages, shipping containers, and the snowplow shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

Snowplow, Truck Mounting, Power Reversible Blade MXU-512/E.

6. NOTES

6.1 Intended use. The MXU-512/E snowplow is intended for mounting on the A/S32U-10, A/S32U-17, A/S32U-35, or A/S32U-36 snow removal equipment truck for use in the removal of snow and slush from airbase runway and taxi areas.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When a snow removal equipment truck will be loaned to the contractor (see 3.14)

c. Location and conditions for preproduction testing (see 4.4.)

d. Applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

Custodian:
Air Force - 11

Reviewers:
Air Force - 82, 85

Preparing activity:
Air Force - 11

Project No. 3830-F116

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM, JAN 68 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC–WPAFB–OCT 67 2M

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